Propositional uniqueness principle for dependent pair types: every element equals its pair of projections

The Univalent Foundations Program

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Editorial note. The counted statement is the author corollary labelled thm:eta-sigma (source lines 1484-1487); the proof span 1488-1491 is the author's proof environment, retained line for line.

Editorial note. The proof applies the path characterization of Sigma types; the retained context carries its statement, so the citation resolves inside the composed document.

Editorial scope. One original result of the HoTT book chapter cha:basics, counted once: the propositional uniqueness principle for dependent pair types. Only the counted corollary and its author proof are retained; the surrounding commentary on introduction and elimination forms remains uncounted in the source.

Allowed standard prerequisites (editorial).

- Dependent pair types with their projections.
- The characterization of paths in dependent pair types as pairs of a base path and a path over it (the retained context).
- Transport along a path with its definitional behaviour on reflexivity, and reflexivity paths.

Theorem 0.0.1. *Suppose that $P : A \to \mathcal{U}$ is a type family over a type A and let $w, w' : \sum_{(x:A)} P(x)$. Then there is an equivalence*

$$(w = w') \simeq \sum_{(p:\mathsf{pr}_1(w)=\mathsf{pr}_1(w'))} p_*(\mathsf{pr}_2(w)) = \mathsf{pr}_2(w').$$

Corollary 0.0.2. *For $z : \sum_{(x:A)} P(x)$, we have $z = (\mathsf{pr}_1(z), \mathsf{pr}_2(z))$.*

Proof. We have $\mathsf{refl}_{\mathsf{pr}_1(z)} : \mathsf{pr}_1(z) = \mathsf{pr}_1(\mathsf{pr}_1(z), \mathsf{pr}_2(z))$, so by Theorem 0.0.1 it will suffice to exhibit a path $(\mathsf{refl}_{\mathsf{pr}_1(z)})_*(\mathsf{pr}_2(z)) = \mathsf{pr}_2(\mathsf{pr}_1(z), \mathsf{pr}_2(z))$. But both sides are judgmentally equal to $\mathsf{pr}_2(z)$. □